Using Network Analysis and Content Analysis to Analyze Participants' Interaction Structures and Inquiry Patterns in an Online Course

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INTRODUCTION

Learning, if not fundamentally, at least is partially a socially-constructed process (Haythornthwaite, 2005; Hodson & Hodson, 1998). Learners construct knowledge not only through their internal thinking process from previous experiences but also through communications and collaborations with others in a social context (Brown & Adler, 2008; Lee, 2014). The rapid development of information and communication technology (ICT) and Internet technology has fostered online interactions and collaborations; accordingly, the concept of online social learning emerges in this digital age. Online social learning often refers to social activities that take place at a distance and in mediated forms (Jones & Steeples, 2002). Online social learning is conceptualized as a collective and participatory social interaction process, including interactions between a learner and another learner, between a learner and an instructor, and between a learner and learning resources (Anderson, 2003; Jones & Steeples, 2002).

THEORETICAL FRAMEWORKS AND RESEARCH METHODS

Social network analysis (SNA) is both a theoretical perspective and a research method that has been widely utilized in analyzing and describing social interactions in online learning settings (de Laat, Lally, Lipponen, & Simons, 2007; Haythornthwaite, 2005). Due to its foundation on relations among social entities, SNA can assist in analyzing, describing, and understanding participants' teaching and learning interaction structures, as well as the flow of information and resources that are exchanged among participants in the online learning (de Laat, Lally, Lipponen, & Simons, 2007). Further, the community of inquiry (CoI) framework (Garrison, Anderson, & Archer, 2000) is adopted to analyze the instructor's and students' inquiry pattern in the online discussions. The three essential elements of the CoI framework are cognitive presence (CP; the extent to which online learners are able to construct meaning and critical thinking through communication), social presence (SP; the ability of individuals to project their personal characteristics into the community), and teaching presence (TP; the instructor designs, plans, and prepares the course, facilitates the discourse and provides direct instruction in the learning process). Recently, Shea et al. have proposed a new CoI construct - learning presence (LP), which refers to "students' proactive use of specific processes, such as setting goals, selecting and deploying strategies, and self-monitoring one's effectiveness to improve their academic achievement" (Shea, et al., 2013). In this study, coding scheme for these four presences are used to identify participants' presences in the online discussions.

RESEARCH PURPOSE, CONTEXT, AND QUESTIONS

The primary purpose of this study is to explore online students' learning interaction structures and inquiry patterns within a completely online graduate-level course offered at a midwestern research university. This online course is focused on exploring the theories of online learning communities and practices of building online learning community, and is part of a certificate program focused on online learning. Twenty graduate students enrolled in this online course; one of the researchers was enrolled as a doctoral student within this course during a 15-week semester in spring 2014. The online course instructor, who has over ten years of online teaching experiences, has put paramount emphasis on improving students' online interaction, participation, and collaboration in online learning environments.

DATA ANALYSIS METHODS AND STRATEGIES

The whole data set include all text-based asynchronous discussions in the Ning forum, the transcript of an online synchronous meeting, as well as transcripts of the instructor's instructional audios, videos, and class agendas. The first phase of this study is using SNA method to analyze the structure and attributes of the student-instructor interaction network in the online course. The data set are manually transcribed into a set of participant-participant data. Butts' network measures (2008) and R package (2008, 2014) are utilized to analyze the student-instructor interaction network. Based on CoI presence coding scheme, quantitative content analysis (QCA) is used to analyze students' inquiry patterns in the online discussions (Riff, Lacy, & Fico, 2014).

RESULTS AND DISCUSSIONS

SNA results: Data is analyzed through the node level and the network level using the SNA method. Table 1 represents the node-level analysis results of the students' and the instructor's indegree, outdegree, betweenness, and closeness centrality. This analysis demonstrates that the instructor, and students E and R have the largest outdegree centrality, indicating that they reply and mention most frequently to others in the discussion activities during the whole semester. The instructor and student J have the largest indegree in the network, indicating that they are replied and mentioned most frequently by others in discussions. Student P has both the lowest indegree and the lowest outdegree in the network, indicating that comparing to other participants, P does not actively participate in the discussions. Figure 1 visually demonstrates the sociograph based on the participant's degree, including indegree and outdegree. The node's size and color, as well as the edge width represent the participant's degree centrality. The instructor, student R, and student C are relatively central in the interaction network. Except for student P, the instructor and other students form a relatively evenly distributed subgroup; P locates far away from the subgroup and D locates slightly far away from the subgroup. In addition, according to Table 1, the student R has the largest betweenness centrality, indicating that student R has the largest influence over the interactions of other individuals in the network. Student E, student R, and the instructor have the largest closeness centrality, indicating they have the best efficiency in spreading information to other participants in the network. Student P has the lowest betweenness and closeness centrality in the network.

Participant	Indegree	Outdegree	Degree	Betweenness	Closeness
A	16	14	30	0.032	0.044
B	15	17	32	0.059	0.050
C	15	17	32	0.037	0.050
D	12	13	25	0.025	0.042
E	15	20	35	0.056	0.057
F	17	18	35	0.048	0.052
G	13	18	31	0.035	0.052
H	16	15	31	0.058	0.046
I	14	17	31	0.042	0.050
J	19	14	33	0.073	0.044
K	17	13	30	0.052	0.042
L	16	11	27	0.025	0.040
M	18	16	34	0.069	0.048
N	15	12	27	0.019	0.041
O	16	18	34	0.043	0.052
P	9	6	15	0.004	0.033
Q	13	17	30	0.050	0.050
R	18	20	38	0.103	0.057
S	16	14	30	0.042	0.044
T	17	16	33	0.041	0.048
Instructor	19	20	39	0.086	0.057

Table 1. Participants' centrality summary

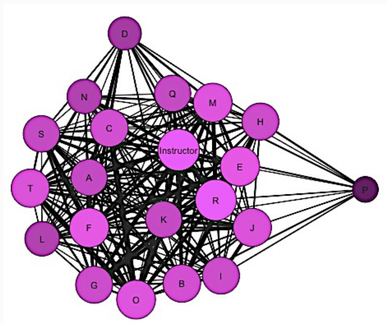


Figure 1. Participants' interaction sociograph

RESULTS AND DISCUSSIONS CONT

QCA results: There are a total of 52 posts and 1249 comments in the discussions. Table 2 demonstrates the frequency and percentage of four presences for each participant in all discussions during the whole semester. The total frequency of four CoI presences is 2202. The results reveal that the occurrences of the four CoI constructs in the discussions are SP (50.5%), CP (37.7%), LP (8.7%) and TP (4.1%). These results indicate that SP frequently occurred in the inquiry process and LP, the new proposed construct, is not frequently demonstrated in the inquiry process during discussions. TP is only demonstrated by the instructor in the discussions.

	SP(%)	CP(%)	LP(%)	TP(%)
A	49(4.4)	32(4.0)	15(7.8)	0(0.0)
B	74(6.7)	55(6.8)	12(6.3)	0(0.0)
C	61(5.5)	38(4.7)	13(6.8)	0(0.0)
D	34(3.1)	24(3.0)	4(2.1)	0(0.0)
E	76(6.8)	54(6.7)	13(6.8)	0(0.0)
F	102(9.2)	69(8.6)	17(8.9)	0(0.0)
G	65(5.8)	51(6.3)	12(6.3)	0(0.0)
H	52(4.7)	40(5.0)	10(5.2)	0(0.0)
I	46(4.1)	39(4.8)	8(4.2)	0(0.0)
J	44(4.0)	44(5.5)	3(1.6)	0(0.0)
K	31(2.8)	31(3.8)	7(3.6)	0(0.0)
L	29(2.6)	27(3.3)	4(2.1)	0(0.0)
M	55(4.9)	35(4.3)	13(6.8)	0(0.0)
N	22(2.0)	27(3.3)	3(1.6)	0(0.0)
O	72(6.5)	51(6.3)	13(6.8)	0(0.0)
P	12(1.1)	12(1.5)	3(1.6)	0(0.0)
Q	29(2.6)	24(3.0)	6(3.1)	0(0.0)
R	108(9.7)	78(9.7)	21(10.9)	0(0.0)
S	39(3.5)	28(3.5)	9(4.7)	0(0.0)
T	44(4.0)	41(5.1)	6(3.1)	0(0.0)
Instructor	68(6.1)	7(0.9)	0(0.0)	91(100)
Total	1112(50.5)	807(36.6)	192(8.7)	91(4.1)
TOTAL: 2202				

Table 2. Participants' CoI presences

IMPLICATIONS

Based on the interaction and participation network analyses, we propose some suggestions for online instructors to build a more interactive online learning community and to provide students a more engaging online learning experience. First, the instructor's interaction and participation play an extremely important role in building an online learning community. The instructors should make efforts to get involved in the online conversations in order to foster students' engagement. Moreover, the instructor should encourage students to participate in more discussions with more peers in a higher frequency. The instructor should also consider provide opportunities for students to take different leadership roles in discussions. In this way, students can form an online learning community, where they co-construct knowledge through actively participating in a variety of online activities with their peers and the instructor.

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